

SIMATS ENGINEERING

C02-ASSESSMENT 3

ITA1407-ETHICAL HACKING

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COURSE OUTCOME COVERED

CO2: Foot printing Tools –: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT4)

Scenario-Based Assessment

Topic 1: Foot printing Tools – Conducting Competitive Intelligence

Q1. (L3 – Apply) – 15 Marks

A cybersecurity consultant has been assigned to perform a passive security assessment for a company named **ABC Technologies**. The consultant must collect publicly available information without directly interacting with the company's servers.

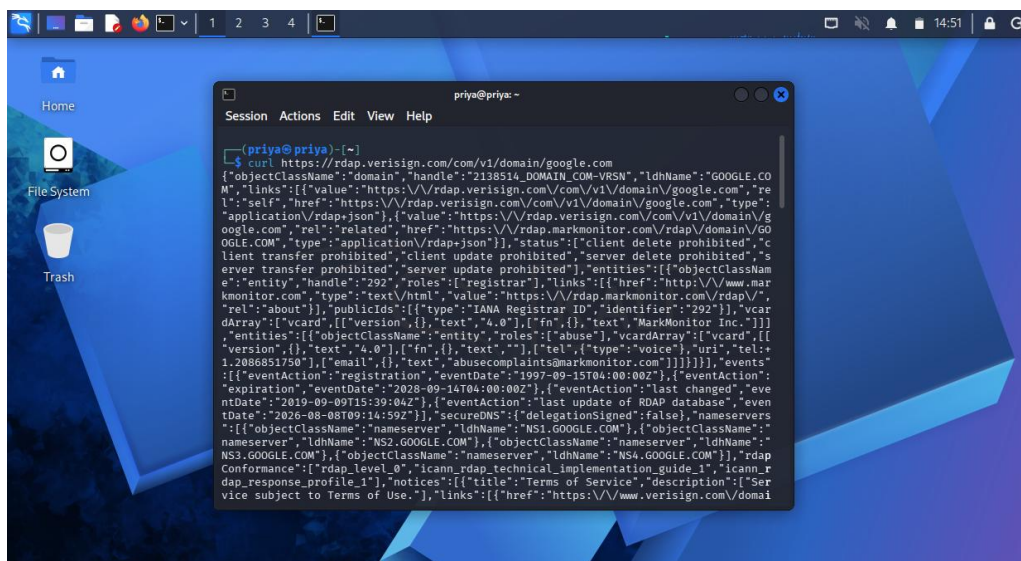
Task: Apply suitable foot printing tools to identify the company's:

- domain registration details,
- employee email addresses,
- technology stack,
- and publicly exposed subdomains.

Explain how each selected tool helps in conducting competitive intelligence.

DOMAIN REGISTRATION DETAILS

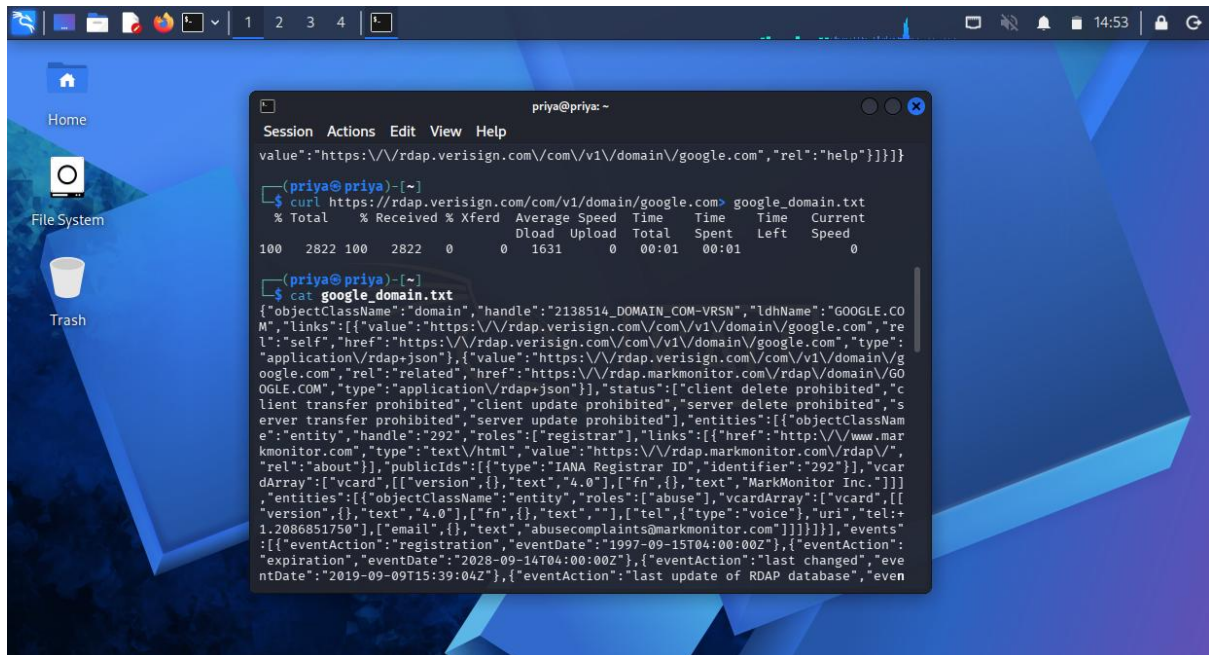
curl <https://rdap.verisign.com/com/v1/domain/google.com>



Command

```
curl https://rdap.verisign.com/com/v1/domain/google.com
```

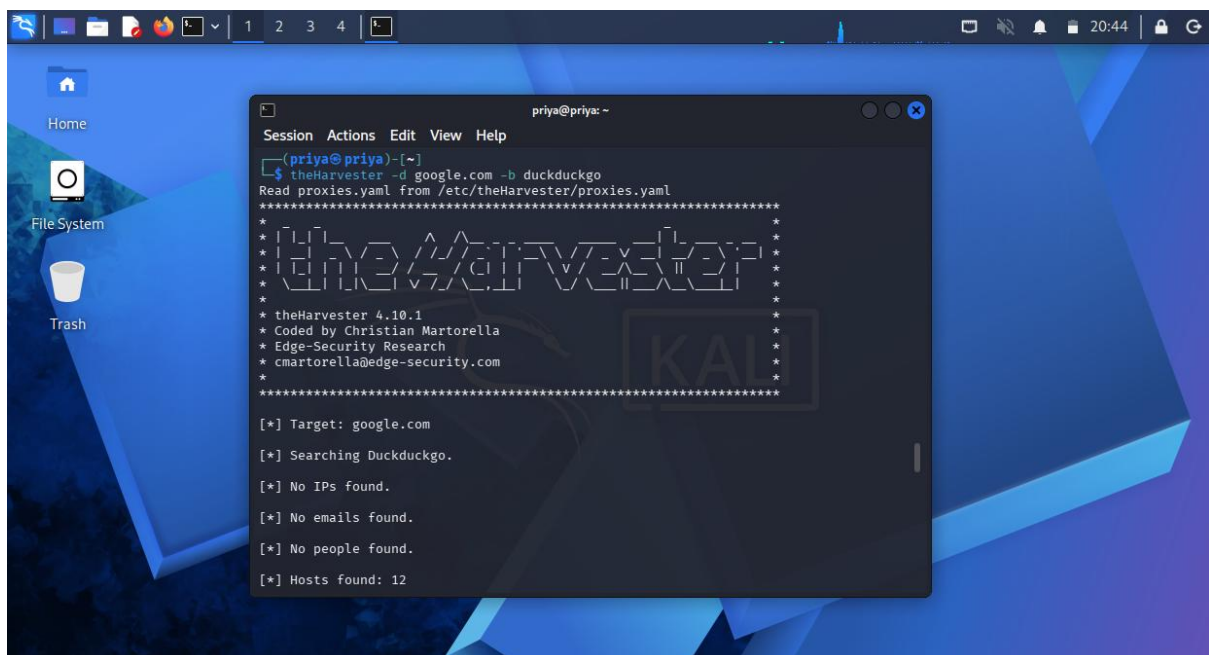
```
cat google_domain.txt
```



EMPLOYEE EMAIL ADDRESSES

Command

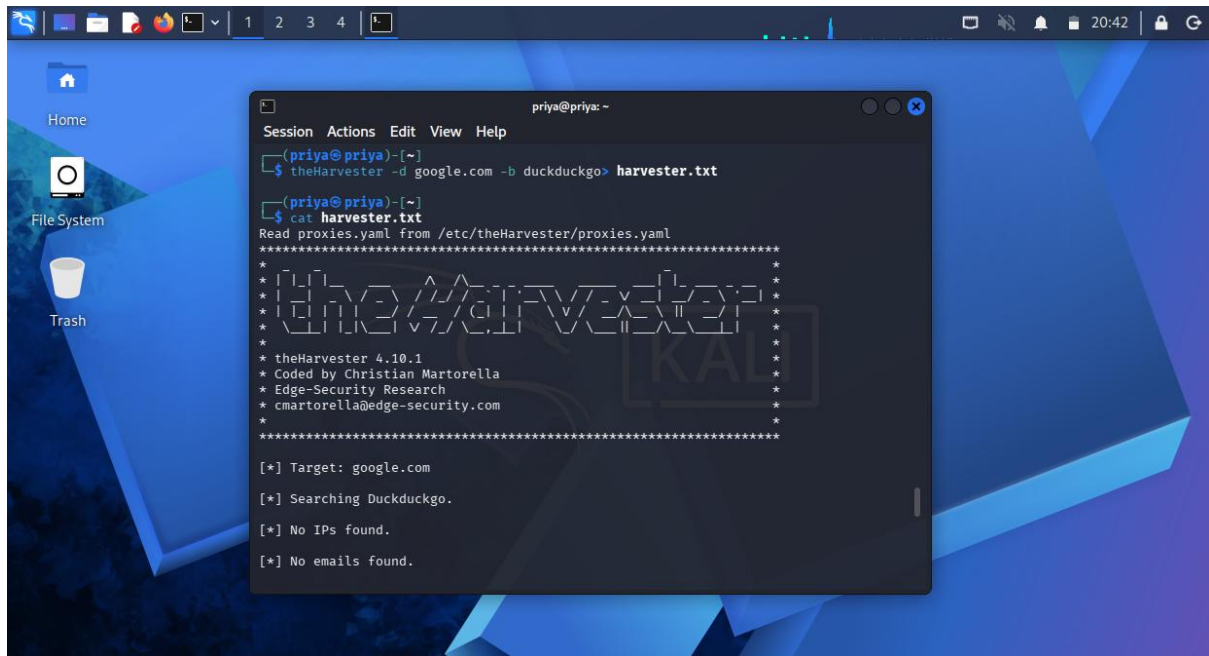
theHarvester -d google.com -b duckduckgo



command

```
cat harvester.txt
```

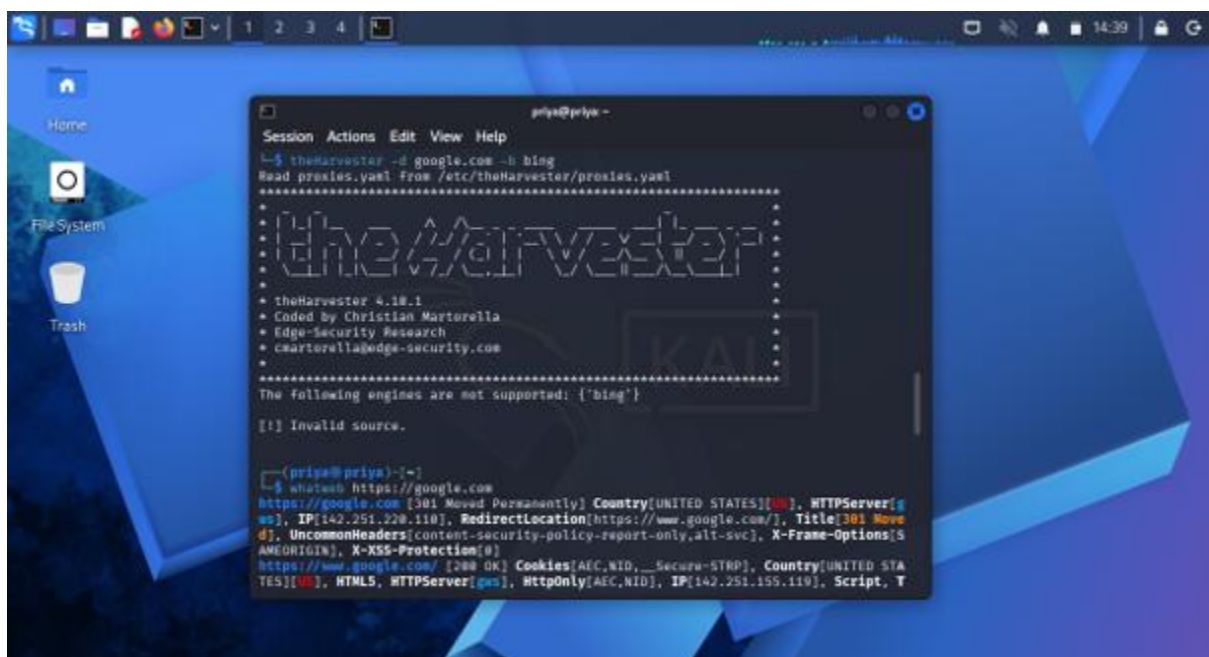
whatweb https://google.com



TECHNOLOGY STACK

Command

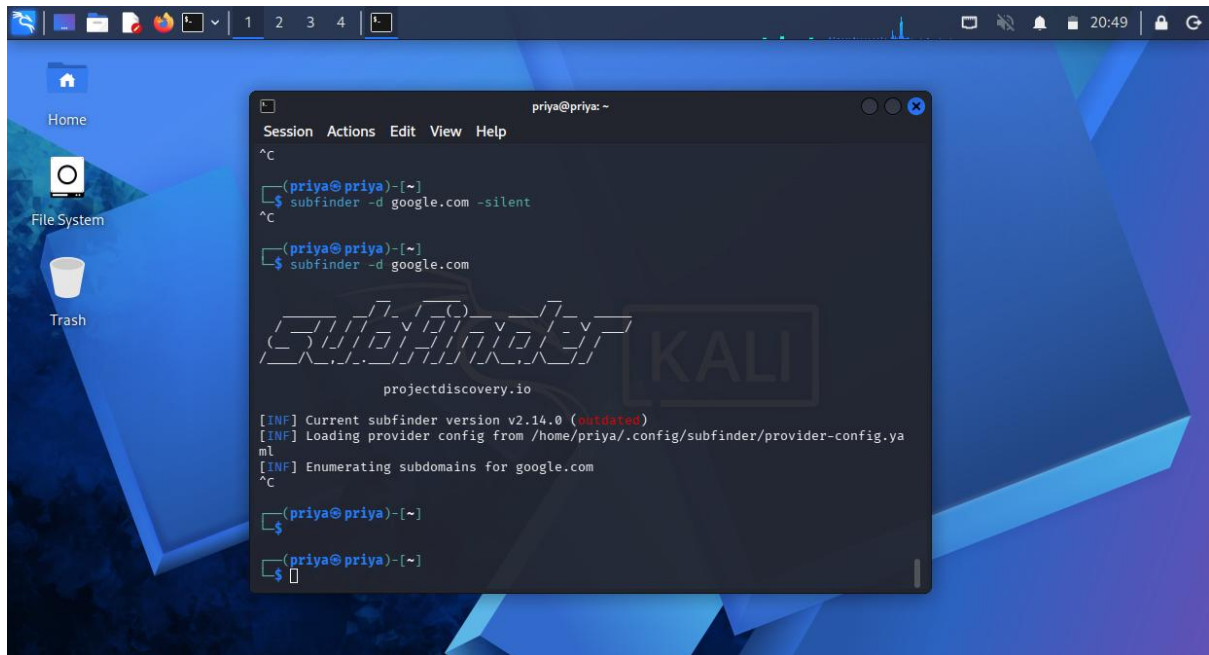
whatweb <https://google.com>



PUBLICLY EXPOSED SUBDOMAINS

Command

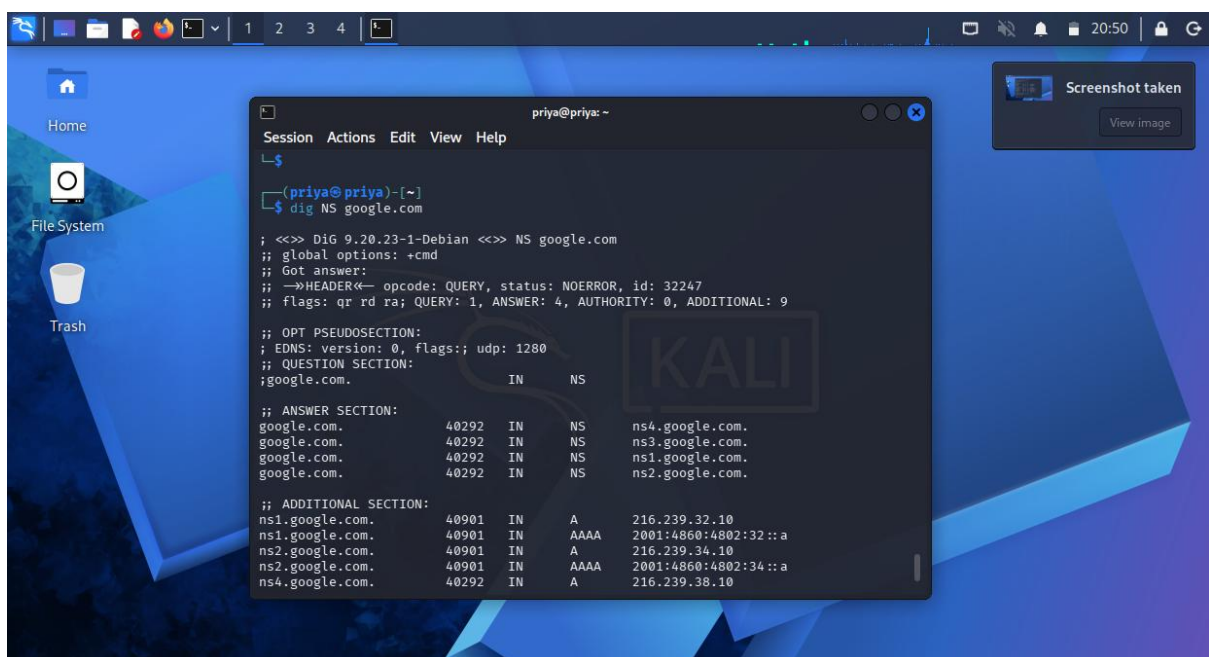
subfinder -d google.com -silent



DNS INFORMATION

Command

dig NS google.com



Topic 2: DNS Zone Transfers

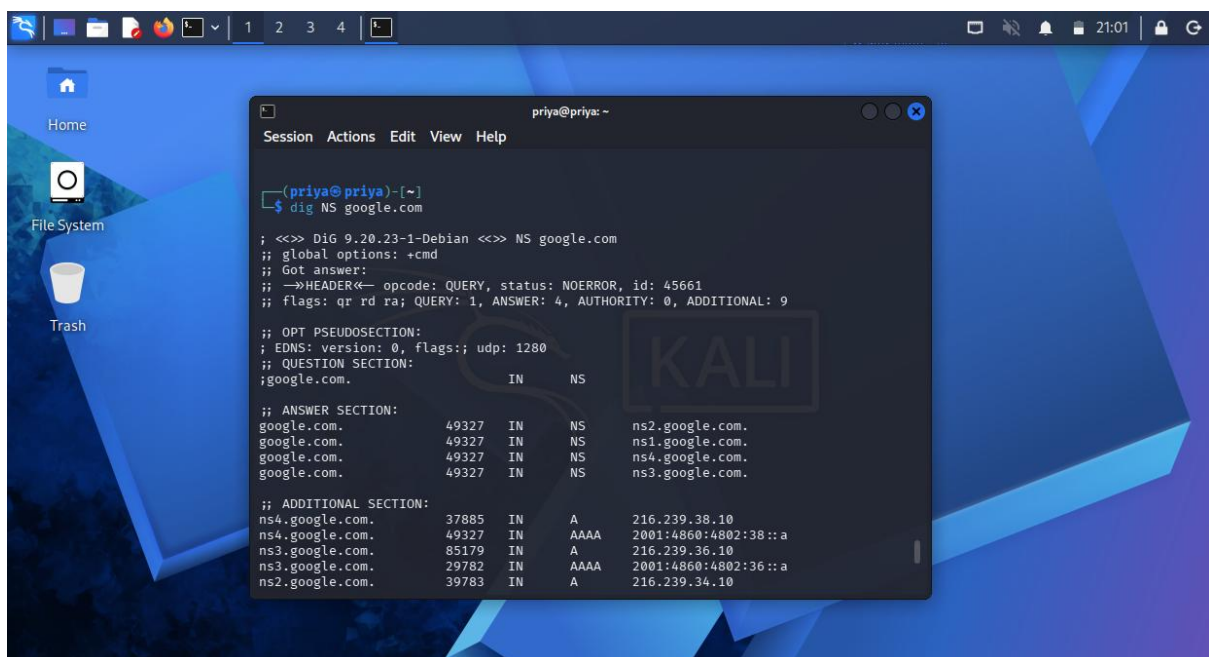
Q2. (L3 – Apply) – 15 Marks

During an authorized penetration test, a security analyst suspects that the target organization's DNS server is misconfigured and may allow unauthorized DNS zone transfers.

Task: Apply the appropriate DNS command or tool to verify whether zone transfers are permitted. If the transfer succeeds, explain how the retrieved DNS records can be used to identify internal hosts, mail servers, and subdomains during reconnaissance.

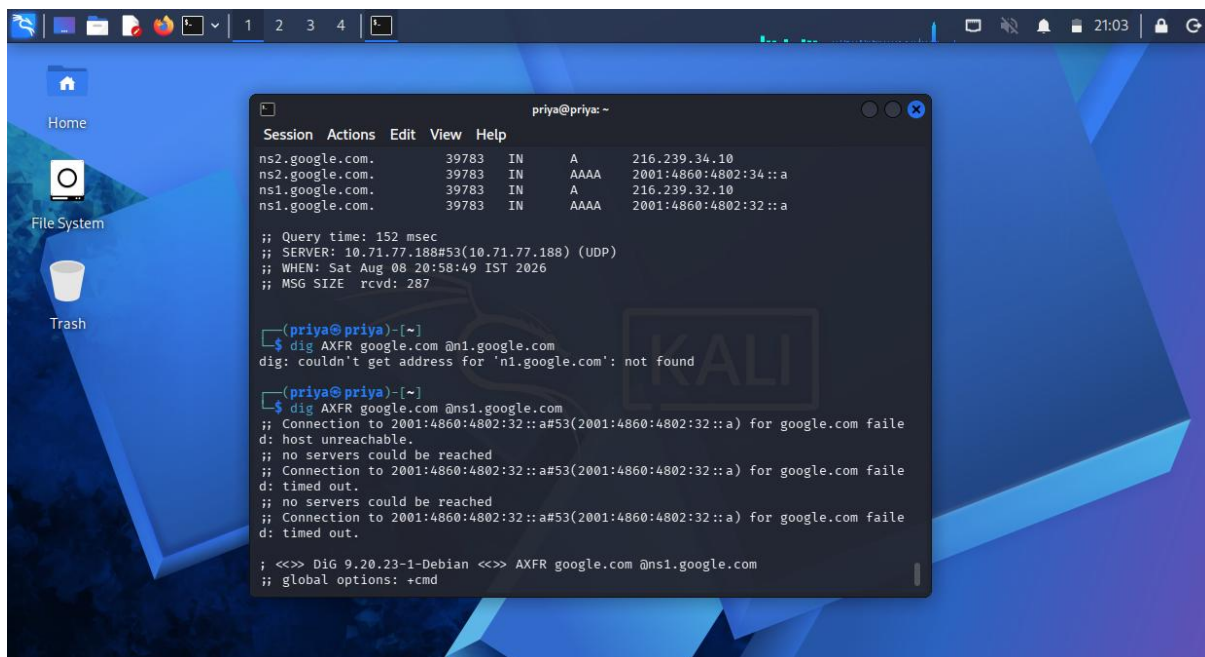
Command

dig NS example.com



```
priya@priya: ~  
Session Actions Edit View Help  
  
(priya@priya)-[~]  
$ dig NS google.com  
  
; <<>> DiG 9.20.23-1-Debian <<>> NS google.com  
;; global options: +cmd  
;; Got answer:  
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 45661  
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 9  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1280  
;; QUESTION SECTION:  
;google.com. IN NS  
  
;; ANSWER SECTION:  
google.com. 49327 IN NS ns2.google.com.  
google.com. 49327 IN NS ns1.google.com.  
google.com. 49327 IN NS ns4.google.com.  
google.com. 49327 IN NS ns3.google.com.  
  
;; ADDITIONAL SECTION:  
ns4.google.com. 37885 IN A 216.239.38.10  
ns4.google.com. 49327 IN AAAA 2001:4860:4802:38::a  
ns3.google.com. 85179 IN A 216.239.36.10  
ns3.google.com. 29782 IN AAAA 2001:4860:4802:36::a  
ns2.google.com. 39783 IN A 216.239.34.10
```

dig AXFR example.com @ns1.example.com

A screenshot of a Kali Linux desktop environment. The desktop background is a blue and purple geometric pattern. On the left side, there are icons for 'Home', 'File System', and 'Trash'. A terminal window is open in the center, displaying the output of a 'dig AXFR' command. The terminal window has a title bar that says 'priya@priya: ~'. The output shows DNS records for ns2.google.com and ns1.google.com, including IP addresses and AAAA records. It also shows query time, server information, and connection status for google.com.

```
priya@priya: ~  
Session Actions Edit View Help  
ns2.google.com. 39783 IN A 216.239.34.10  
ns2.google.com. 39783 IN AAAA 2001:4860:4802:34::a  
ns1.google.com. 39783 IN A 216.239.32.10  
ns1.google.com. 39783 IN AAAA 2001:4860:4802:32::a  
  
;; Query time: 152 msec  
;; SERVER: 10.71.77.188#53(10.71.77.188) (UDP)  
;; WHEN: Sat Aug 08 20:58:49 IST 2026  
;; MSG SIZE rcvd: 287  
  
(priya@priya)~  
$ dig AXFR google.com @ns1.google.com  
dig: couldn't get address for 'ns1.google.com': not found  
  
(priya@priya)~  
$ dig AXFR google.com @ns1.google.com  
;; Connection to 2001:4860:4802:32::a#53(2001:4860:4802:32::a) for google.com failed: host unreachable.  
;; no servers could be reached  
;; Connection to 2001:4860:4802:32::a#53(2001:4860:4802:32::a) for google.com failed: timed out.  
;; no servers could be reached  
;; Connection to 2001:4860:4802:32::a#53(2001:4860:4802:32::a) for google.com failed: timed out.  
  
; <<>> Dig 9.20.23-1-Debian <<>> AXFR google.com @ns1.google.com  
;; global options: +cmd
```

Topic 3: Introduction to Social Engineering – Social Engineering Attacks and Countermeasures

Q3. (L3 – Apply) – 20 Marks

Employees of an organization receive an email claiming to be from the IT department, asking them to verify their login credentials through a provided link because of an urgent security update.

Task: Identify the type of social engineering attack and apply suitable countermeasures that employees should follow before responding to the email. Explain how these actions help prevent credential theft.

COMMAND

Dig google.com

